TIP

Being able to link data about orders and sales to operational data from sensors allows you to start to answer questions from customers about the quality of goods you have delivered to them for example.

FINANCE, PROCUREMENT, AND BUSINESS INTELLIGENCE

Integrating with financial and procurement systems lets a digital twin factor costs into operational decisions. Understanding financial data such as payment terms and cash conversion cycles can enable, for example, recommendations for inventory levels that will minimize working capital while maintaining operational performance. An autonomous digital twin may automatically place an order for a replacement part from a supplier based on operational data from sensors.

Many organizations have existing investments in business intelligence and analytics tools that contain pre-processed metrics, KPIs, and business rules that digital twins can leverage rather than recreating. These platforms may already compute equipment utilization rates, energy efficiency metrics, or maintenance cost analyses that provide valuable inputs for digital twin algorithms and simulations.

4.2.3 The convergence of IT and OT

Digital twins that are accessed by end users to inspect, simulate, and predict a physical system's behaviors most often execute within the IT network, but rely on data from OT systems. Consider using Google Maps to look at traffic congestion on your route home. You would typically use a web browser to access the map view. The data that controls the congestion coloring within the map may be sourced from a traffic management system such as the *Sydney coordinated adaptive traffic system (SCATS)* which is an OT system deployed in many cities across the world to manage and control traffic signals. Given the threats posed to IT systems (think of viruses and malware that you have encountered on personal computers), these cannot be connected directly to an OT system. Imagine a virus from your home computer infecting the city traffic control system.

CHALLENGES IN OT DATA INTEGRATION

Working with OT data presents unique challenges in simply accessing the data, with industrial systems often operating on physically isolated ('air gapped') networks with strict security requirements to protect physical assets. Many organizations use data diodes or unidirectional gateways that only allow data to flow out of OT networks, making real-time querying impossible and forcing digital twins to work with replicated data.